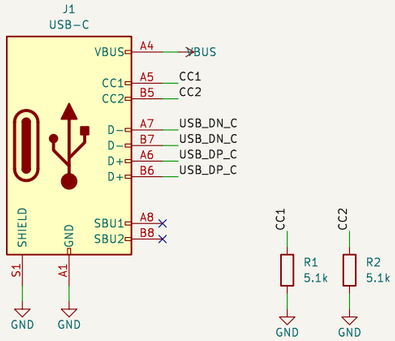
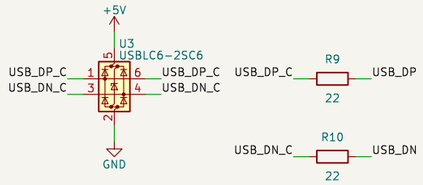
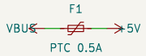
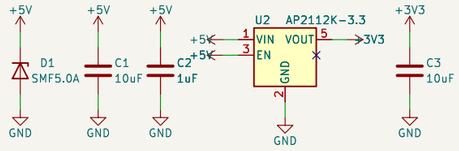
1. USB-C (USB 2.0 data + 5 V)





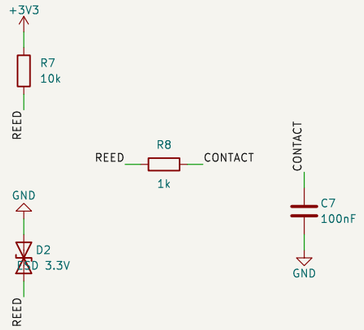
2. Protection + 3.3 V LDO





3. Reed switch input (GPIO4, closed = LOW)

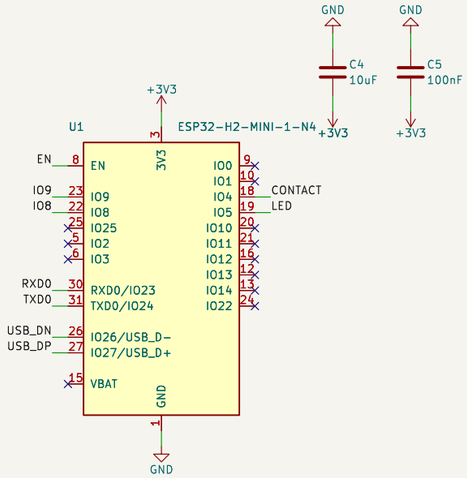




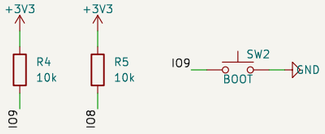
Power flags



4. ESP32-H2-MINI-1-N4

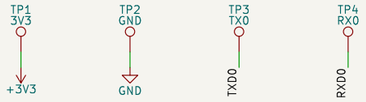


5. Reset, boot / factory reset, strapping



Status LED (GPIO5)

6. Test pads (bottom side): UART0 log / recovery



NOTES
1. PCB 26 x 46 mm, 2 layers, 1.6 mm. Every SMD part on the top side; J2 (screw terminal) hand soldered.
2. U1 at the top edge, antenna to the edge: no copper, tracks or vias on either layer under the antenna area.
3. Pins = firmware defaults (main/Kconfig.projbuild): reed switch GPIO4, BOOT / factory reset GPIO9 (hold 6 s), status LED GPIO5 active high (enable CONFIG_STATUS_LED_ENABLE). USB Serial/JTAG on GPIO26/27.
4. Reed input: R7 10k pull-up at the terminal (about 0.3 mA wetting current), R8 1k + C7 100nF filter, D2 ESD. The firmware also enables the GPIO4 internal pull-up and debounces in software.
5. Flashing and logs over USB-C (USB Serial/JTAG). Download mode by hand: hold BOOT (GPIO9 low, GPIO8 pulled high by R5) and press RESET. USB-C: CC1/CC2 5.1k each (5 V sink), PTC + TVS on VBUS, USBLC6 on D+/D-.
6. VBAT (pin 15) is tied to 3V3 inside the module (datasheet v1.6): not connected here.
7. Unused GPIO: IO0-IO3, IO10-IO14, IO22, IO25 (IO25 strapping, floating = default). H1/H2: M2 holes, PCB only.

USB-C 5 V + USB Serial/JTAG, reed switch on GPIO4 (screw terminal)
2-layer 1.6 mm, 26 x 46 mm, all SMD parts on the top side

Sheet: /
File: h2_contact.kicad_sch

Title: h2-contact: ESP32-H2 Matter over Thread door/window contact sensor

Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1